

RRC Communication Protocol with the Host Computer Analysis

1. Description

This chapter mainly analyzes the communication protocol between the RRC development board and the host computer (ROS package), laying the foundation for subsequent learning about the host computer.

2. Communication Protocol

Common instruction examples can be found in: Common Communication Instruction Examples

Instructions are written in hexadecimal. If you are not familiar with the calculation method, you can refer to: Using a calculator tool for hexadecimal conversion. For converting negative numbers and floating-point numbers to hexadecimal, please search for online tutorials.

Instruction format:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	(uint8_t) Function	(uint8_t) Length	Data	(uint8_t) CRC

- Frame Header: Continuous reception of 0xAA, 0x55 indicates the arrival of a data packet.
- Function Code: Used to indicate the purpose of an information frame.
- Data Length: Number of parameters.

- Parameters: Additional control information other than function instructions.
- Checksum: Verifies the correctness of the data using CRC checksum method (CRC calculation of Function, Length, and Data values, taking the lower 8 bits).

2.1 User-Initiated Data Transmission to the Control Board

Section

The development board already has a dedicated UART to USB circuit, so communication only requires connecting the UART3 port to the host computer using a data cable.

1. LED control: Instruction name **PACKET_FUNC_LED**, value 1

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_LED	7	Parameter 1: (uint8_t) led_id Parameter 2: (uint16_t) light-on duration(ms) Parameter 3: (uint16_t) light-off duration(ms) Parameter 4: (uint16_t) number of cycles	CRC

Example:

① **Control the LED to blink 5 times, with each cycle having the light on for 100ms and off for 100ms:**

Frame	Function Code	Data	Parameter	Checksum
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Header		Length		
0xAA 0x55	PACKET_FUNC_LED	7	Parameter 1: 0x01 (1) Parameter 2: 0x64 0x00 (100) Parameter 3: 0x64 0x00 (100) Parameter 4: 0x05 0x00 (5)	CRC

Note: This is little-endian mode. For example, the uint32_t value 5 (decimal) is written as 0x00 0x05. In little-endian mode, the least significant byte comes first, so when sending data, it should be written as 0x05 0x00.

② Control the LED to blink 10 times, with each cycle having the light on for 500ms and off for 300ms

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_LED	7	Parameter 1: 0x01 (1) Parameter 2: 0xF4 0x01 (500) Parameter 3: 0x2C 0x01 (300) Parameter 4: 0x0A 0x00 (10)	CRC

2. Buzzer control: Instruction name PACKET_FUNC_BUZZER, value

2

Frame Header	Function Code	Data Length	Parameter	Checksum
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0xAA 0x55	PACKET_FUNC_BUZZER	8	Parameter 1: (uint16_t)frequency (Hz) Parameter 2: (uint16_t) beeping duration(ms) Parameter 3 : (uint16_t) no-beeping duration(ms) Parameter 4: (uint16_t) number of cycles	CRC
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Example:

①Control the buzzer to sound 5 times, with a frequency of 1400Hz, each lasting for 100ms followed by a 100ms pause

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUZZER	8	Parameter 1 : 0x78 0x05 (1400) Parameter 2 : 0x64 0x00 (100) Parameter 3 : 0x64 0x00 (100) Parameter 4 : 0x05 0x00 (5)	CRC

②Control the buzzer to sound 10 times, with a frequency of 1000Hz, each lasting for 500ms followed by a 300ms pause.

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUZZER	8	Parameter 1 : 0xE8	CRC

			0x03 (1000) Parameter 2 : 0xF4 0x01 (500) Parameter 3 : 0x2C 0x01 (300) Parameter 4 : 0x0A 0x00 (10)	
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3. Encoder motor control: Instruction name **PACKET_FUNC_MOTOR**, value 3

(1) Control Single Motor

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	6	Parameter 1: (uint8_t) 0x00 (subcommand) Parameter 2: (uint8_t) motor_id Parameter 3: (float) speed value (positive and negative)	CRC

(Note: float type is 4 bytes)

Example: Control motor 1 to rotate at -1r/s speed:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	6	Parameter 1: 0x00 Parameter 2: 0x01 Parameter 3 : 0x00 0x00 0x80 0xBF (-1)	CRC

(2) Control Multiple Motors

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	5N+2	Parameter 1: (uint8_t) 0x01 (subcommand) Parameter 2: (uint8_t) motor quantity Parameter 3 : (uint8_t) motor_id_1 Parameter 4: (float) speed_1 Parameter 2N+1: (uint8_t) motor_id_N Parameter 2N+2: (float) speed_N (Format reference parameters 3, 4)	CRC

(Note: float type is 4 bytes)

Example:

Control motor 1 and motor 2 to rotate at speeds of -1r/s and 2r/s respectively:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	12	Parameter 1 : 0x01 (subcommand) Parameter 2: 0x02 Parameter 3: 0x01 Parameter 4 : 0x00	CRC

			0x00 0x80 0xBF (-1) Parameter 5: 0x02 Parameter 6 : 0x00 0x00 0x00 0x40 (+2)	
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(3) Control Single Motor to Stop

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	2	Parameter 1: (uint8_t) 0x02 (subcommand) Parameter 2 : (uint8_t) motor_id	CRC

Example:

Control motor 1 to stop:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	2	Parameter 1 : 0x02 (subcommand) Parameter 2: 0x01	CRC

(4) Control several motors to stop:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	2	Parameter 1: (uint8_t) 0x03 (subcommand) Parameter 2: (uint8_t) motor mask	CRC

Example:

Control motor 0 and motor 2 to stop:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_MOTOR	2	Parameter 1: 0x03 (mode) Parameter 2 : 0x05 (Binary 00000101)	CRC

4. PWM servo control: instruction name

PACKET_FUNC_PWM_SERVO, value 4

(1) Control several PWM servos

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	3N+3	Parameter 1 : (uint8_t) 0x01 (subcommand) Parameter 2: (uint16_t) motion time(ms) Parameter 3: (uint8_t)servo_id_1 Parameter 4: (uint16_t) pulse width Parameter 2N+1 : (uint8_t) servo_id_N Parameter 2N+2: (uint16_t)pulse	CRC

			width	
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Example: Control servo 1 and servo 2 to rotate to 90° and 180° respectively, corresponding to pulse widths of 1500 and 2500, within 2 seconds:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	9	Parameter 1 : 0x01 (subcommand) Parameter 2: 0xD0 0x07 (2000) Parameter 3: 0x01 (1) Parameter 4: 0xDC 0x05 (1500) Parameter 5: 0x02 (2) Parameter 6: 0xC4 0x09 (2500)	CRC

(2) Control single PWM servo

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	6	Parameter 1 : (uint8_t) 0x03 (subcommand) Parameter 2: (uint16_t) motion time(ms) Parameter 3: (uint8_t) servo_id Parameter 4: (uint16_t) pulse width	CRC

The pulse width ranges from [500, 2500], corresponding to [0°, 180°].

Example:

Control servo 1 to rotate to the 90° position within 1 second, with a pulse width of 1500:

Frame Header	Function Code	Data Length	Parameter	Check sum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	6	Parameter 1 : 0x03 (subcommand) Parameter 2: 0xE8 0x03 (1000) Parameter 3: 0x01 (1) Parameter 4: 0xDC 0x05 (1500)	CRC

(3) Read PWM servo position

Frame Header	Function Code	Data Length	Parameter	Check sum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	2	Parameter 1 : (uint8_t) 0x05 (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the command to read the position, the control board will send the position data of the corresponding PWM servo with the ID to the host computer."

(4) Set PWM Servo Deviation

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	3	Parameter 1 : (uint8_t) 0x07 (subcommand) Parameter 2: (uint8_t) servo_id Parameter 3: (uint8_t) deviation parameter (Convert to int8_t type, valid range is -100 to +100)	CRC

Example:

Set the offset of servo 2 to +10

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	3	Parameter 1 : 0x07 (subcommand) Parameter 2: 0x02 (2) Parameter 3: 0x0A (10)	CRC

(5) Read PWM Servo Deviation

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_PWM_SERVO	2	Parameter 1 : (uint8_t) 0x09	CRC

	0		(subcommand)	
			Parameter 2: (uint8_t) servo_id	

After sending the command to read the position, the control board will send the offset data of the corresponding PWM servo with the ID to the host computer.

5. Serial bus servo control: Instruction name

PACKET_FUNC_BUS_SERVO, value 5

(1) Control serial bus servo to move

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	3N+4	Parameter 1: (uint8_t) 0x01 (subcommand) Parameter 2: (uint16_t) motion time(ms) Parameter 3: (uint8_t) servo quantity Parameter 4: (uint8_t) servo_id_1 Parameter 5: (uint16_t) Pulse width Parameter 2N+1: (uint8_t) servo_id_N Parameter 2N+2: (uint16_t) Pulse width	CRC

Example:

Control bus servo 1 and servo 2 to rotate to 200° and 240° respectively, corresponding to pulse widths of 833 and 1000, within 1 second:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	10	Parameter 1 : 0x01 (subcommand) Parameter 2: 0xE8 0x03 (1000) Parameter 3: 0x02 (2) Parameter 4: 0x01 (1) Parameter 5: 0x41 0x03 (833) Parameter 4: 0x02 (2) Parameter 5: 0xE8 0x03 (1000)	CRC

(2) Read position command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	2	Parameter 1 : (uint8_t) 0x05 (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the command to read the position, the control board will send the position data of the corresponding bus servo with the ID to the host computer.

(3) Read the input voltage command

Frame Header	Function Code	Data Length	Parameter	Checksum

0xAA 0x55	PACKET_FUNC_BUS_SERV O	2	Parameter 1 : (uint8_t) 0x07 (subcommand) Parameter 2: (uint8_t) servo_id	CRC
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After sending the command to read the input voltage, the control board will send the input voltage data of the corresponding bus servo with the ID to the host computer.

(4) Read temperature command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERV O	2	Parameter 1 : (uint8_t) 0x09 (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the command to read the temperature, the control board will send the temperature data of the corresponding bus servo with the ID to the host computer.

(5) Motor power-off command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERV O	2	Parameter 1 : (uint8_t) 0x0B (subcommand) Parameter2: (uint8_t) servo_id	CRC

After sending the command to unload power, the control board will unload the power of the corresponding bus servo with the ID.

(6) Motor power-on command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	2	Parameter 1 : (uint8_t) 0x0C (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the command to load power, the control board will load power to the corresponding bus servo with the ID.

(7) ID writing command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	3	Parameter1: (uint8_t) 0x10 (subcommand) Parameter 2: (uint8_t) servo_id Parameter3: (uint8_t) save_id	CRC

After sending the ID write command, the control board will overwrite the ID number of the corresponding bus servo.

(8) ID reading command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	2	Parameter1 : (uint8_t) 0x12 (subcommand) Parameter2 : (uint8_t) 0xFE (Broadcast Query)	CRC

After sending the ID read command, the control board will upload the ID number of the bus servo to the host computer.

Note: There can only be one bus servo here. Otherwise, multiple bus servos returning data simultaneously will cause bus conflicts.

(9) Deviation adjusting command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	3	Parameter 1 : (uint8_t) 0x20 (subcommand) Parameter 2: (uint8_t) servo_id Parameter 3: (uint8_t) deviation adjustment value	CRC

After sending the offset adjustment command, the control board will set the offset value of the corresponding bus servo with the value of parameter 3.

(10) Deviation reading command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	2	Parameter 1 : (uint8_t) 0x22 (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the offset read command, the control board will read the offset value of the corresponding bus servo and upload it to the host computer.

(11) Deviation saving command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	2	Parameter 1 : (uint8_t) 0x24 (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the offset save command, the control board will enable the corresponding bus servo with the ID to save its current offset value.

(12) Position limit setting command

Frame Header	Function Code	Data Length	Parameter	Checksum

		h		
0xAA 0x55	PACKET_FUNC_BUS_SERV O	6	Parameter 1 : (uint8_t) 0x30 (subcommand) Parameter 2: (uint8_t) servo_id Parameter 3: (uint16_t) lower limit Parameter 4: (uint16_t) higher limit	CRC

The numerical range of the limit is [0, 1000].

After sending the position limit setting command, the control board will set the limit parameters for the corresponding ID of the bus servo.

(13) Position limit reading command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERV O	2	Parameter 1 : (uint8_t) 0x32 (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the position limit read command, the control board will upload the limit parameters of the corresponding bus servo to the host computer.

(14) Voltage Limit Setting Command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA	PACKET_FUNC_BUS_SERV	6	Parameter 1 : (uint8_t) 0x34	CRC

0x55	O		(subcommand) Parameter 2: (uint8_t) servo_id Parameter 3: (uint16_t) lower limit Parameter 4: (uint16_t) higher limit	
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The low voltage limit value should be greater than 4500, and the high voltage limit value should be less than 14000.

After sending the voltage limit setting command, the control board will set the voltage limit parameters for the corresponding ID of the bus servo.

(15) Voltage limit reading command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERV O	2	Parameter 1 : (uint8_t) 0x36 (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the voltage limit read command, the control board will upload the voltage limit parameters of the corresponding bus servo to the host computer.

(16) Temperature limit setting command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERV O	3	Parameter 1 : (uint8_t) 0x38 (subcommand)	CRC

			Parameter 2: (uint8_t) servo_id Parameter 3 : (uint8_t) high temperature threshold	
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The parameter value for the high-temperature threshold should be less than 100.

After sending the temperature limit setting command, the control board will set the high-temperature threshold parameter for the corresponding ID of the bus servo.

(17) Temperature limit reading command

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	2	Parameter 1 : (uint8_t) 0x3A (subcommand) Parameter 2: (uint8_t) servo_id	CRC

After sending the temperature limit setting command, the control board will upload the high-temperature threshold parameter of the corresponding bus servo to the host computer.

2.2 Data sent by the control board to the user

1. Bus Servo Data Upload: Command name

PACKET_FUNC_BUS_SERVO, value 5

(1) Serial bus servo position upload

Frame	Function Code	Data	Parameter	Che
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Header		Length		checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	5	Parameter 1: (uint8_t) servo_id Parameter 2 : (uint8_t) 0x05 (subcommand) Parameter 3: (int8_t) Was the read successful? (0: success; -1: fail) Parameter 4 : (int16_t) servo position	CRC

When a read command is received, the corresponding servo's position parameter is read and uploaded to the host computer.

Example:

The angle of servo 5 is uploaded as 30°, corresponding to a pulse width of 833:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	5	Parameter 1: 0x05 (5) Parameter 2 : 0x05 (subcommand) Parameter 3: 0x00 (0) Parameter 4 : 0x41 0x03 (833)	CRC

(2) Bus Servo Voltage Upload

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	5	Parameter 1: (uint8_t) servo_id Parameter 2: (uint8_t) 0x07 (subcommand) Parameter 3: (int8_t) Was the read successful? (0: success; -1: fail) Parameter 4: (uint16_t) servo voltage	CRC

When a read command is received, the voltage parameter of the corresponding servo is read and uploaded to the host computer.

(3) Bus Servo Temperature Upload

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	4	Parameter 1: (uint8_t) servo_id Parameter 2: (uint8_t) 0x09 (subcommand) Parameter 3: (int8_t) Was the reading successful (0: success; -1: fail) Parameter 4: (uint8_t) servo temperature	CRC

When a read command is received, the temperature parameter of the

corresponding servo is read and uploaded to the host computer.

(4) Serial Bus Servo ID Upload

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	4	Parameter 1 : (uint8_t) 0xFE (broadcast command) Parameter 2 : (uint8_t) 0x12 (subcommand) Parameter 3 : (int8_t) Was the reading successful (0: success; -1: fail) Parameter 4 : (uint8_t) read servo_id	CRC

When a read command is received, the ID number of the servo is read and uploaded to the host computer.

(5) Serial Bus Servo Deviation Parameter Upload

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	4	Parameter 1 : (uint8_t) 0xFE (previous broadcast command) Parameter 2 : (uint8_t) 0x22 (subcommand)	CRC

			Parameter 3: (int8_t) Was the reading successful (0: success; -1: fail) Parameter 4: (uint8_t) servo deviation	
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When a read command is received, the deviation parameter of the corresponding servo is read and uploaded to the host computer.

(6) Bus Servo Position Limit Parameter Upload

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERVO	7	Parameter 1: (uint8_t) servo_id Parameter 2: (uint8_t) 0x32 (subcommand) Parameter 3: (int8_t) Was the reading successful (0: success; -1: fail) Parameter 4: (uint16_t) lower limit Parameter 5: (uint16_t) higher limit	CRC

When a read command is received, the position limit parameters of the corresponding servo are read and uploaded to the host computer.

(7) Serial Bus Servo Voltage Limit Parameter Upload

Frame	Function Code	Data Length	Parameter	Checksum
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Header		th		m
0xAA 0x55	PACKET_FUNC_BUS_SERV 0	7	Parameter 1: (uint8_t) servo_id Parameter 2 : (uint8_t) 0x36 (subcommand) Parameter 3: (int8_t) Was the reading successful (0: success; -1: fail) Parameter 4: (uint16_t) lower limit Parameter 5: (uint16_t) higher limit	CRC

When a read command is received, the voltage limit parameters of the corresponding servo are read and uploaded to the host computer.

(8) Bus Servo Temperature Limit Parameter Upload

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_BUS_SERV 0	4	Parameter 1: (uint8_t) servo_id Parameter 2 : (uint8_t) 0x3A (subcommand) Parameter 3: (int8_t) Was the reading successful	CRC

			(0: success; -1: fail) Parameter 4: (uint8_t)temperature limit	
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When a read command is received, the temperature limit parameters of the corresponding servo are read and uploaded to the host computer.

2. Upload Button Message: Instruction Name **PACKET_FUNC_KEY**, Value 6

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_KEY	2	Parameter 1 : (uint8_t) button_id Parameter 2 : (uint8_t) button event	CRC

Partial button event callback values are as follows:

BUTTON_EVENT_PRESSED = 0x01, /* Button pressed /

BUTTON_EVENT_LONGPRESS = 0x02, / Button long pressed /

BUTTON_EVENT_CLICK = 0x20, / Button clicked /

BUTTON_EVENT_DOUBLE_CLICK = 0x40, / Button double-clicked */

Example:

Send a message indicating that button 1 is pressed:

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_KEY	2	Parameter 1: 0x01 Parameter 2: 0x01	CRC

3. Upload IMU Data: Command Name PACKET_FUNC_IMU, Value 7

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_IMU	24	Parameter 1 : (float) accel_x data Parameter 2 : (float) accel_y data Parameter 3 : (float) accel_z data Parameter 4 : (float) gyro_x data Parameter 5 : (float) gyro_y data Parameter 6 : (float) gyro_z data	CRC

4. Upload Gamepad Data: Command Name PACKET_FUNC_GAMEPAD, Value 8

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_IMU	7	Parameter 1 : (uint16_t) button Parameter 2: (uint8_t) hat value Parameter 3: (int8_t) left joystick x value Parameter 4: (int8_t) left	CRC

			joystick y value Parameter 5: (int8_t) right joystick x value Parameter 6: (int8_t) right joystick y value	
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5. Upload SBUS model aircraft remote control data: Command name

PACKET_FUNC_SBUS, value 9

Frame Header	Function Code	Data Length	Parameter	Checksum
0xAA 0x55	PACKET_FUNC_SBUS	36	Parameter 1 : (int16_t[16])16 channel values Parameter 2 : (uint8_t)ch17 Parameter 3 : (uint8_t)ch18 Parameter 4 : (uint8_t)signal_loss Parameter 5 : (uint8_t)fail_safe	CRC

The data for all 16 channels of the model aircraft's joysticks and switches are contained in parameter 1.

6. PWM servo control: Command name **PACKET_FUNC_PWM_SERVO**, value 4

(1) PWM servo position upload

Frame	Function Code	Data	Parameter	Checks
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Header		Length		um
0xAA 0x55	PACKET_FUNC_PWM_SERV O	4	Parameter 1: (uint8_t)servo_id Parameter 2 : (uint8_t)0x05 (subcommand) Parameter 3 : (uint16_t)pulse width	CRC

Upon receiving the read command, read the pulse width of the corresponding servo and upload it to the host computer.

(2) PWM servo deviation upload

Frame Header	Function Code	Data Length	Parameter	Checks um
0xAA 0x55	PACKET_FUNC_PWM_SERV O	3	Parameter 1: (uint8_t)servo_id Parameter 2 : (uint8_t)0x05 (subcommand) Parameter 3 : (int8_t)servo deviation	CRC

Upon receiving the read command, read the pulse width of the corresponding servo and upload it to the host computer.

Common Communication Protocol Examples

Preparation:

Open the serial port assistant, connect via type-C to UART2, set the baud rate to 1000000, and select HEX (hexadecimal transmission).

Examples:

1. Control the buzzer to sound 5 times, at a frequency of 1400Hz, each time for 100ms and stop for 100ms:

```
AA 55 02 08 78 05 64 00 64 00 05 00 F0
```

2. Control the buzzer to sound 10 times, at a frequency of 1000Hz, each time for 500ms and stop for 300ms:

```
AA 55 02 08 E8 03 F4 01 2C 01 0A 00 8B
```

3. Control the LED light to flash 10 times, each time on for 500ms and off for 300ms:

```
AA 55 01 07 01 F4 01 2C 01 0A 00 04
```

4. Control the LED light to flash 5 times, each time on for 100ms and off for 100ms:

```
AA 55 01 07 01 64 00 64 00 05 00 37
```

5. Single motor control movement: Control motor 1 to rotate at -1r/s speed:

```
AA 55 03 06 00 01 00 00 80 BF DA
```

6. Control motor 1 to stop:

```
AA 55 03 02 02 01 08
```

7. Control motor 1 and motor 2 to rotate at -1r/s and 2r/s respectively:

```
AA 55 03 0C 01 02 01 00 00 80 BF 02 00 00 00 40 FB
```

8. Control motors 0 and 2 to stop:

```
AA 55 03 02 03 05 AD
```

9. Control servo 1 and servo 2 to rotate to 90° and 180° respectively, corresponding to pulse widths of 1500, 2500, 2s:

```
AA 55 03 09 01 D0 07 01 DC 05 02 C4 09 83
```

10. Control PWM servo 1 to move to 1000:

```
AA 55 04 06 03 e8 03 01 e8 03 e4
```

11. Set PWM servo deviation: Set the deviation of servo 2 to +10:

```
AA 55 03 03 07 02 0A 53
```

12. Control bus servo rotation: Control bus servo No. 1 and No. 2 to 200° and 240° respectively, the corresponding pulse widths are 833 and 1000, and it takes 1s:

```
AA 55 05 0A 01 E8 03 02 01 41 03 02 E8 03 9F
```

13. Control the bus servo to rotate: Control bus servo No. 1 and No. 2 to rotate to 0°, the corresponding pulse width is 0, and the time is 1s:

```
AA 55 05 0A 01 E8 03 02 01 00 00 02 00 00 D2
```

14. Control the bus servo with ID 1 to power off:

```
AA 55 05 02 0B 01 B3
```

15. Control the bus servo with ID 1 to power on:

```
AA 55 05 02 0C 01 DD
```

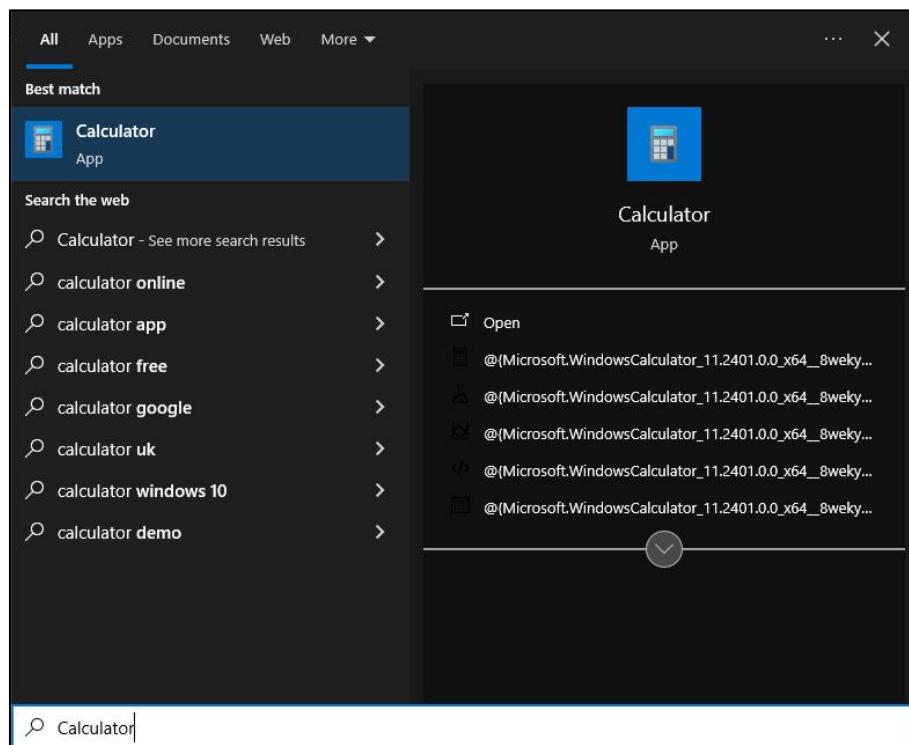
16. Change the ID of the bus servo with ID 1 to 2:

```
AA 55 05 03 10 01 02 68
```

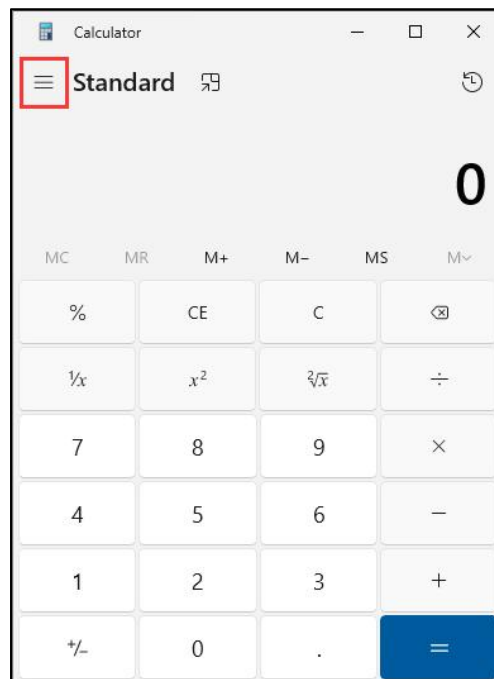
Expansion—Using the Calculator Tool for Base Conversion

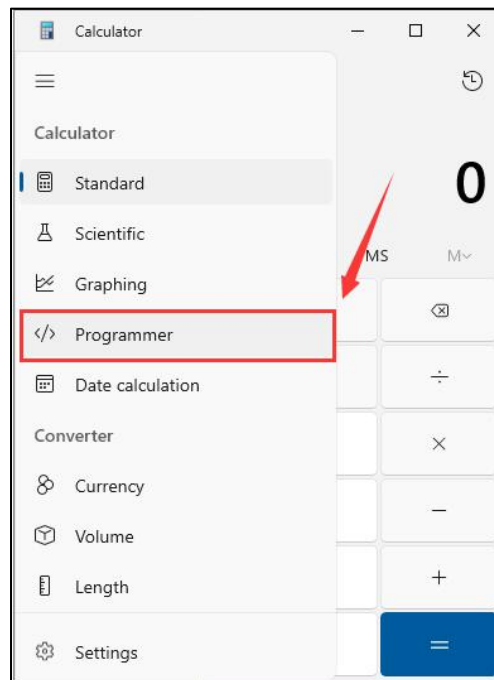
Using the Windows Calculator for Base Conversion:

①: Open the Start menu, search for Calculator, and open it.



②: Click on the Open Navigation button in the top left corner, and select Programmer.





③: Four bases are available for selection. Choose one base, input a value, and the equivalent values in the other three bases will be displayed.

